SUMAN AZAD

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EDUCATION

B.Sc. in Electrical and Electronic Engineering

Bangladesh University of Engineering and Technology (BUET)

Sep 2015 — Dhaka, Bangladesh

CGPA: **3.37 out of 4.00**, Major: **Power**
Relevant coursework: Power Electronics; Energy Conversion I, II, III; Power System I, II; Power System Protection; Power System Operation and Control; Control System I; Numerical Methods

RESEARCH EXPERIENCE

SoC-Aware Droop Control of a PV–Battery Islanded AC Microgrid

Aug 2025 – Apr 2026 — Bangladesh

- Proposed an **SoC-adaptive droop mechanism** for heterogeneous BESS units in AC microgrids to address unequal energy depletion—an issue not handled by conventional droop methods.
- Developed a detailed **9-bus microgrid model in MATLAB/Simulink** with heterogeneous GFM/GFL inverter architectures, incorporating realistic line impedances to evaluate power-sharing performance under non-ideal network conditions.
- Designed adaptive droop coefficients to dynamically regulate active power dispatch based on battery SoC for balanced utilization across asymmetric storage units.
- Preliminary validation of the core GFM/GFL droop architecture on a simplified lossless 5-bus model prototype (130 kW PV + 110 kW BESS): frequency nadir **49.67 Hz under 80% load step**, battery sharing error **< 5%**, secondary frequency restoration to **49.999 Hz within 2.4 s**.
- Code & technical report of this prototype: github.com/.../droop-controlled-pv-battery-microgrid
- Submitted a manuscript on IET-GTD based on this research work which is currently under review. Manuscript ID - GTD-2026-05-0284.

Undergraduate thesis on "A new topology for loadability of power transmission lines."

Supervisor: Dr. Kazi Mujibur Rahman, Professor, BUET

Jul 2014 - Sep 2015 — BUET

- Proposed a topology to increase the loadability of conventional power transmission lines using cyclo-converter based frequency converters.
- Demonstrated that reducing transmission frequency increases line loadability via reduction of line reactance ($X = 2\pi f L$), validated analytically through the power transfer equation $P_e = \frac{E_1 E_2}{X} \sin\delta$.
- Simulation results confirmed the theory: reducing frequency by a factor of 3 ($50\,Hz \rightarrow 16.67\,Hz$) increased maximum transferable power from $20.6\,MW$ to $56.5\,MW$ — a $2.74\times$ improvement in loadability.

PUBLICATIONS

Journal Articles

- **S. Azad**, A. Chowdhury, "Performance Assessment of SoC-Adaptive Frequency Droop Control in a Heterogeneous Islanded AC Microgrid," *IET Generation, Transmission & Distribution*, submitted May 2026, **under review**. (Manuscript ID: GTD-2026-05-0284)

Conference Papers

- **S. Azad**, D. Shome, F. Faisal, "Multi-Horizon Short-Term Electricity Demand Forecasting for Bangladesh's National Grid: A Comparative Study of Deep Learning and Machine Learning Approaches," in Proc. *2026 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON-2026)*, **presented**, August 2026.
- S. Roy, K. M. Galib, **S. Azad**, D. Shome, S. S. Saeed, and M. T. Siraj, "Clean Energy to the People: How Solar PV Became One of the Most Affordable Energy Sources," in Proc. *3rd Int. Conf. Mechanical Engineering and Applied Sciences (ICMEAS-2025)*, Dhaka, Bangladesh, July 2025.

RESEARCH INTERESTS

- Power systems analysis — system resiliency, power flow, security, and stability
- Power electronics — grid-tied inverter control, filter design, ac-ac/dc-dc converter design, inverter and rectifier design
- Renewable energy integration — PV systems, MPPT control, and inverter-dominated power systems

- Microgrid control — GFM and GFL inverter coordination, droop control, and inertia estimation methodology
- Application of machine learning for power systems resiliency, security, stability, and control

SKILLS

- **Programming:** Python (NumPy, Pandas, TensorFlow/Keras, Scikit-learn), C#, PHP.
- **Simulation & Modeling:** MATLAB, Simulink, PSAF, PSCAD.
- **Database management systems:** MYSQL, MSSQL.
- **Other skills:** Technical report writing, design documentation review, electrical system commissioning, web & software development.

TEST SCORES

GRE General Test

Score: 316
(Quant: 165, Verbal: 151, AWA: 3.5)

15th Sep 2026

IELTS - Academic

Overall: 7.5 (Listening - 8.5, Reading - 8.5, Writing - 7.5, Speaking - 6.0)

1st Feb 2026

JOB EXPERIENCE

Senior Assistant Manager - Electrical

Nuclear Power Plant Company Bangladesh Ltd

Feb 2018 – Ongoing — Pabna, Bangladesh

- Commissioned 10kV and 0.4kV switchgear, 24/10kV auxiliary transformers, remote terminal units of the 2400MW nuclear facility.
- Reviewed and approved unit auxiliary power supply system working drawings, 400 kV and 230 kV GIS drawings for the power evacuation and back-feed power supply network.
- Reviewed design documentations, working documentations, and commissioning documents of the power units; review and approval of design changes according to international normative documents.
- Testing of circuit breakers, transformers and protective relays of 10kV and 0.4kV switchgear.
- Commissioned electrical motors, power supply cabinets, cable lines, limit switches, transport locks, sliding gates of 360/32/10 ton polar crane and transport portal crane in the reactor building of Rooppur NPP.
- Commissioned the emergency power supply system consisting DC boards, batteries, inverters, rectifiers, and diesel generators of safety channel I & II of Rooppur power units.

Assistant Engineer

Power Grid Company Bangladesh Ltd

Nov 2017 - Feb 2018 — Dhaka, Bangladesh

- Operated and maintained 132/33kV grid substation with 2x80/120MVA transformers and a 165km transmission line.
- Performed CT, VT, and transformer testing; participated in transmission line inspection and protection coordination.
- Participated in substation manpower management, store management, procurement, and civil works.

AWARDS

University Merit Scholarship
For achieving 21st position out of 10,000 candidates appearing in the admission test of BUET at 2009-10 session.

Education board scholarship
Awarded for excellent results in the high school board scholarship exam at 2004-2005 session.

TRAININGS & COURSES

Process Management Training Program - Electrical Shop

Rosatom Technical Academy, Novovoronezh, Russia

Jan 2019 - Aug 2019 — Certificate link

- Basic training program for Electrical Shop Engineers of Rooppur Power Plant.
- Theoretical, practical and on-job training on auxiliary systems of power plants, GIS for power evacuation, maintenance of major electrical equipment, fault analysis and protection schemes of power systems.

Machine Learning A-Z: AI, Python & R

Udemy.com

May 2025 - Jul 2025 Certificate link

- **Topics covered:** Data pre-processing; Machine learning techniques: Regression, Classification, Clustering; Associate rule learning, Reinforcement learning, Natural language processing; Deep learning, Artificial neural network, Convolutional neural network; Dimensional reduction techniques, Model selection and Boosting techniques

NOTABLE PROJECTS

Grid-Connected Solar PV System Simulation, Modeling, and THD Analysis

Tools: MATLAB, Simulink

Github Project Link

Modeled and simulated a grid-connected photovoltaic system with MPPT control, inverter synchronization, and harmonic analysis under dynamic irradiance conditions. Simulation results — **MPPT tracking efficiency 99%**, stable DC-link voltage during irradiance changes, proper grid synchronization, grid current **THD below 4%**.
Report PDF: docs.google.com/../file

24-Hour Power Demand Forecasting with Machine Learning Models

Tools: Python, Neural Network, XGBoost, Ensemble

Github Project Link

This project applies multiple regression-based machine learning models to forecast short-term (next 24 hours) electricity demand for the Bangladesh Power Grid. The models are trained on real demand data from Power Grid Company of Bangladesh (PGCB) and evaluated based on prediction accuracy. Among different models, **LightGBM** achieved the **lowest RMSE (625.94)**, making it the best-performing model in this study.

Combined Cycle Power Plant Output Prediction Using ANN

Tools: Python, Tensorflow, Keras, Pandas, Numpy

Github Project Link

This project uses an Artificial Neural Network (ANN) to predict the net hourly electrical energy output (MW) of a combined cycle power plant. By using key environmental and operational features, the model provides accurate power output predictions achieving an **R^2 score of 93.56%** on the test data.